

README.md

README for “Equilibrium in the Market for Public School Teachers: District Wage Strategies and Teacher Comparative Advantage”

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This README documents the replication package for the Journal of Political Economy paper “Equilibrium in the Market for Public School Teachers: District Wage Strategies and Teacher Comparative Advantage.”

Overview

The code in this replication package constructs teacher-, student-, district-, model-estimation, and simulation datasets and generates the tables and figures listed below and displayed in the paper and Online Appendix.

The original analysis relies on confidential Wisconsin Department of Public Instruction (WDPI) data and cannot be redistributed in this public archive. The public replication package hence includes fixed synthetic input data in `synthetic_data/raw/`. These files are fictitious data created to match the file names, folder structure, variable names, storage formats, and key relationships needed by the replication code. They are not sampled from, perturbed from, or otherwise derived from confidential WDPI records. Results produced from these synthetic data verify the code path and file dependencies but do not reproduce the published numerical estimates.

Public replicators should run `code/master_synth.do`. It reads the synthetic raw inputs from `synthetic_data/`, constructs intermediate/model/simulation files locally, and writes synthetic-run exhibits to `out_synth/`. The confidential-data driver is `code/master.do`; it is not needed for the public synthetic-data replication. In this package, `out_synth/` is the output-simulated folder: it contains the expected tables and figures from the public simulated-data run. `out/` is reserved for actual restricted-data outputs from `code/master.do` and will not be generated in the public replication.

Data Availability and Provenance Statements

Statement About Rights

The confidential WDPI staff, roster, demographic, and test-score data used in the paper cover 2006–2016. The authors first obtained access in November 2017 under a Data User Agreement with WDPI. WDPI did not provide a formal version number, extract identifier, or release number for the delivery. These restricted data are not distributed with this replication package.

Researchers who want access to the confidential data can submit a request to WDPI at:

<https://dpi.wi.gov/wise/data-requests/confidential-data-application>

There is no application or access fee. Access requires IRB approval and a signed Data User Agreement, is subject to WDPI review and approval, and may take several months. The Data User Agreement requires the confidential data to be destroyed after the project is completed. Redistribution of confidential WDPI microdata is not permitted.

In June 2026, the JPE Data Editor approved the use of synthetic data for the public replication in place of the confidential WDPI files. `real_data/` contains shareable auxiliary inputs that are either constructed by the authors or publicly available. These files are included in the public replication archive subject to the applicable source terms and agreements. `synthetic_data/raw/` contains the fictitious WDPI-style inputs distributed for the public replication.

Summary of Availability

Data source	Location expected by code	Publicly provided?	Notes
WDPI staff, roster, and student test-score files	File names listed below	No	Confidential/restricted data covering 2006–2016. First accessed in November 2017; no formal version or release number was provided. Access is free but requires IRB approval, a signed DUA, and WDPI approval, which may take several months.
Auxiliary district crosswalks, distance files, shapefiles, elections, and NCES/ACS extracts	<code>real_data/</code>	Yes	Shareable files constructed by the authors or publicly available.
Synthetic public replication data	<code>synthetic_data/raw/</code>	Yes	Fixed fictitious WDPI-style raw files distributed with the public package. Together with the included auxiliary inputs in <code>real_data/</code> , these are sufficient to run <code>code/master_synth.do</code> . The simulation script is also included at <code>synthetic_data/code/make_simulated_data.do</code> so public replicators can inspect or regenerate the fictitious inputs if needed.
Generated intermediate/model/simulation data	<code>intermediate_data/</code> , <code>model_data/</code> , <code>simulation_data/</code>	No	Created locally when <code>code/master_synth.do</code> runs. They should not be treated as source data.
Expected simulated-run outputs	<code>out_synth/</code>	Yes	Tables and figures generated from the simulated inputs. This is the package’s output-simulated folder.
Actual restricted-data outputs	<code>out/</code>	No	Reserved for outputs from the confidential-data workflow. Public replicators will not generate this folder from the public simulated data.

Preservation and Licensing

The final public replication package will be preserved in the JPE/Harvard Dataverse. The confidential WDPI data cannot be deposited there and cannot be retained indefinitely: the governing Data User Agreement requires their destruction after the project is completed.

The authors' code in `code/` and `synthetic_data/code/` is released under the MIT License. The authors' original synthetic data and documentation are released under the Creative Commons Attribution 4.0 International License (CC BY 4.0). Third-party auxiliary inputs are not relicensed and remain subject to their original source terms. See LICENSE for details.

Raw Datasets Used in the Analysis

The following list contains all the raw files used in the analysis. WDPI staff, roster, and student files are confidential and not provided. Files under `real_data/` are shareable auxiliary inputs constructed by the authors or publicly available.

Raw input	Source/provenance from local evidence
Linked_Staffing_Data_10202015.csv	Confidential WDPI linked staffing data.
06staff.txt through 14staff.txt	Confidential WDPI staff files.
15staff.csv, 16staff.csv	Confidential WDPI staff files.
student_demographics.csv	Confidential WDPI student demographics.
testscores.csv	Confidential WDPI student test scores.
real_data/cesa.dta	CESA-district crosswalk manually created by authors using WDPI CESA information .
real_data/wisconsin_elections.xlsx	Daily Kos Elections statewide election results and county-congressional-district crosswalk, accessed in 2022; see <code>real_data/readme_elections.rtf</code> and the formal citations below.
real_data/crosswalk_district_county.dta	District-county crosswalk prepared by authors using the Opportunity Insights county-commuting-zone-state crosswalk documented below.
real_data/crosswalk_census_state_id.csv	Census district ID to state district ID crosswalk prepared by authors using NCES EDGE School District Boundaries . The exact source vintage was not recorded.
real_data/district_county_cz.dta	District-county-commuting-zone crosswalk prepared by authors using the Opportunity Insights county-commuting-zone-state crosswalk documented below.
real_data/district_distance_sdid.dta	District-to-district distance matrix constructed by authors using Google Maps in September 2022.
real_data/nces_acs_2013–2017/EDGE_Export_101244429626/DP03.6_001_USSchoolDistrictAll_101244423682.txt	NCES/ACS 2013–2017 school-district income extract. The original download date was not recorded.
real_data/nces_acs_2013–2017/EDGE_Export_101244639981/DP03.8_001_USSchoolDistrictAll_101244637329.txt	NCES/ACS 2013–2017 school-district poverty extract. The original download date was not recorded.
real_data/nces_acs_2013–2017/EDGE_Export_1012447445/DP02.7_001_USSchoolDistrictAll_101244740697.txt	NCES/ACS 2013–2017 school-district education extract. The original download date was not recorded.
real_data/nces_acs_2013–2017/EDGE_Export_101253245819/DP05.1_001_USSchoolDistrictAll_101253240998.txt	NCES/ACS 2013–2017 school-district age extract. The original download date was not recorded.
real_data/district_shapefiles/WI_School_Districts.*	WDPI <i>School Districts, Wisconsin</i> , edition 8.0, revised June 26, 2024 and valid for the 2024–2025 school year, plus converted Stata files. Downloaded in 2025 from the WDPI GIS Maps page.
real_data/cz_id.csv	District-commuting-zone crosswalk prepared by authors using the Opportunity Insights county-commuting-zone-state crosswalk documented below.
real_data/district_id_tag.dta and real_data/district_id_tag_validation.dta	Data-district-id to model-district-id crosswalks prepared by authors.

Synthetic Public Data

The public archive replaces the confidential WDPI staff and student inputs with fixed synthetic files under `synthetic_data/raw/`. These files are fictitious records with the same broad structure expected by the build scripts.

Synthetic input	Unit of observation	Notes
synthetic_data/raw/staff/06staff.txt through 14staff.txt	Staff district-school-position-year	Fixed-width fictitious staff files matching the legacy WDPI staff-file layout.
synthetic_data/raw/staff/15staff.csv, 16staff.csv	Staff district-school-position-year	CSV fictitious staff files matching the later WDPI staff-file layout.
synthetic_data/raw/staff/temp.dta	Staff-year linkage support	Fictitious support copy checked by the synthetic-data validator; it is not read by the production synthetic master.
synthetic_data/raw/teacher_data_linked/Linked_Staffing_Data_10202015.csv	Staff-year linkage support	Fictitious linked-staffing support file required by <code>oldteacher_roster_build.do</code> .
synthetic_data/raw/testscores_indlev/student_demographics.csv	Student-year	Fictitious student demographics.
synthetic_data/raw/testscores_indlev/dem.dta	Student-year	Stata-format support copy checked by the synthetic-data validator; the production build creates its working demographics file from the CSV.
synthetic_data/raw/testscores_indlev/testscores.csv	Student-subject-year	Fictitious math, reading, and ELA test-score records.

The synthetic staff files are designed so staff IDs overlap heavily across years, staff observations are district-school-position-year records, most person-years have one district-school-position, and most one-position person-years are teachers. Synthetic test-score and demographic files share fictitious student IDs and school years, with student records repeated across years. The synthetic data also include nonempty cells needed by value-added, teacher-choice, event-time, subgroup, wage-schedule, and map/figure code.

The synthetic inputs are distributed as final fictitious raw files. The simulation script used to generate them is included for transparency at `synthetic_data/code/make_simulated_data.do`. That script creates the fictitious WDPI-style raw inputs from scratch, does not read confidential records, calls `synthetic_data/code/02_make_synthetic_data.do`, and then runs `synthetic_data/code/03_validate_synthetic_structure.do` to check the simulated data's structure and key relationships. Running it overwrites files under `synthetic_data/raw/` and refreshes the public manifest/notes. Additional documentation is in `synthetic_data/documentation/synthetic_data_notes.md` and `synthetic_data/documentation/synthetic_data_manifest.csv`; both files are included in the public archive.

Repository Structure

The main folders and files in the replication folder are:

Path	Purpose
README.md	This replication guide.
LICENSE	License terms for the authors' code, synthetic data, and documentation, with third-party materials excluded from the license grant.
code/	Stata, R, and Fortran code for data construction, analysis, model, and exhibit generation.
code/master_synth.do	Public Stata driver using the fixed synthetic WDPI-style inputs and writing exhibits to <code>out_synth/</code> .
code/master.do	Main Stata driver. Defines globals and calls build/exhibit scripts.
code/produce_exhibits.do	Secondary Stata driver. After the master and Fortran results are produced, calls exhibit scripts to complete the remaining outputs.
code/build/	Stata scripts that construct roster, staff, teacher panel, test-score, value-added, wage-schedule, model-estimation, and simulation input files.
code/descriptives/	Stata scripts for descriptive tables, figures, appendix robustness tables, and map/figure construction.
code/auxiliary_regressions/	Stata scripts for auxiliary teacher-choice and wage regressions and robustness variants.
code/fortran_model/	Fortran code to run the model and all counterfactual and other modeling analyses
code/fortran_model/run_all_local.sh	Portable compiler/runner for all 31 Fortran scenarios on macOS and Unix-like systems. It compiles only unless the replicator explicitly supplies <code>--run</code> .
code/Figure_B9/	Figure B9 Stata code and R simulation helper for the identification illustration.
intermediate_data/	Working intermediate Stata datasets. Current files may be derived from restricted inputs.
model_data/	Model-estimation inputs and CSV exports generated by <code>build_model_dataset.do</code> and related scripts. Current files may be derived from restricted inputs.
simulation_data/	Pre-Act 10 simulation input files generated by <code>build_model_dataset_pre2011.do</code> . Current files may be derived from restricted inputs.
out/	Actual restricted-data exhibit outputs from <code>code/master.do</code> . This folder is not generated by the public simulated-data replication.
out_synth/	Expected simulated-run outputs from <code>code/master_synth.do</code> using the public synthetic data. This is the package's output-simulated folder.
real_data/	Shareable auxiliary inputs that are either constructed by the authors or publicly available.
synthetic_data/raw/	Fixed public synthetic inputs that replace the confidential WDPI staff and student files.
synthetic_data/code/make_simulated_data.do	Public Stata script that regenerates and validates the simulated WDPI-style raw inputs from scratch.
synthetic_data/documentation/	Public manifest and notes describing the simulated raw inputs. Private profiling logs and Stage 1 planning notes are not required for the public archive.

Computational Requirements

Software

Software	Requirement/evidence
Stata 19	Stata/SE or Stata/MP can run the public synthetic workflow. The confidential driver requests a larger variable ceiling and therefore requires Stata/MP. The public workflow was tested with StataNow/MP 19.5 for Apple Silicon, 21 May 2025 revision. <code>code/master.do</code> and <code>code/master_synth.do</code> verify or install the required SSC packages before running. An Internet connection is needed on the first run unless they are already installed.
Stata packages	The tested SSC versions are <code>estout</code> 3.33, <code>esttab</code> 2.1.4, <code>ftools</code> 2.50.0, <code>reghdfe</code> 6.13.1, <code>coefplot</code> 1.8.8, <code>binscatter</code> 7.02, <code>carryforward</code> 4.5, <code>spmap</code> 1.3.2, and <code>sepscatter</code> 1.1.0. The drivers install current SSC releases rather than enforcing these exact versions.
R	Required only for <code>code/Figure_B9/cutoff_prefs_id_example_graph.R</code> . The analysis used R 4.2.3. The helper was also tested with R 4.5.1, <code>haven</code> 2.5.5, <code>extRemes</code> 2.2.1, and <code>MASS</code> 7.3-65. It creates the configured user library when needed and installs missing packages from CRAN.
Fortran and MPI	The model was originally run with GNU Fortran (GCC) 11.5.0 20240719 (Red Hat 11.5.0-14) and <code>mpiexec</code> (OpenRTE) 4.1.1. The distributed code was also compiled successfully on Apple Silicon with GNU Fortran 16.1.0 and Open MPI 5.0.9.
BLAS/LAPACK	BLAS and LAPACK are required by the Fortran model. The portable runner uses Apple's Accelerate framework on macOS and <code>-lblas -llapack</code> on other Unix-like systems. <code>FORTTRAN_LINK_FLAGS</code> can override this choice.
Shell/OS	Command-line examples use a Unix-like shell. The portable Fortran runner requires Bash. Start Stata in the repository root as instructed below. The Stata drivers locate the replication root from Stata's current working directory; no machine-specific path edits are required.

Tested Hardware, Runtime, and Storage

The public Stata workflow and local compilation checks were run on a MacBook Pro (Mac16,6) with an Apple M4 Max processor (16 CPU cores), 128 GB of memory, and macOS 26.5.2. The installed Stata/MP license used two Stata cores.

In two clean validation runs on this machine, optional synthetic-input regeneration took approximately 1 hour 31 minutes to 1 hour 43 minutes, and `code/master_synth.do` took approximately 2 hours 43 minutes to 2 hours 53 minutes. The Figure B9 R helper took less than three seconds. The complete 31-scenario Fortran stage took approximately 12 hours 29 minutes with eight MPI ranks, and the final exhibit step took less than five seconds.

Using the included synthetic raw inputs, the required Stata, R, Fortran, and final-exhibit pipeline therefore took approximately 15 hours 13 minutes. Including optional synthetic-input regeneration increased the total to approximately 16 hours 44 minutes. Replicators should allow an overnight or full-day run; hardware and MPI configuration can materially affect runtime.

The complete working tree occupied approximately 3.8 GB after the Stata workflow generated its intermediate and model data. Replicators should allow at least 5 GB of free disk space. More space may be needed for Fortran logs, platform-specific build files, or additional validation copies.

Controlled Randomness

`code/master_synth.do` sets both `set seed 12345` and `set sortseed 12345` to control Stata randomness. The synthetic master also normalizes opened/closed timestamps in Stata table logs so repeated public runs produce the same substantive synthetic tables and figures. The simulated-data generator `synthetic_data/code/make_simulated_data.do` sets both `set seed 20260613` and `set sortseed 20260613`. The generator also normalizes Stata `.dta` timestamps and unused padding in the two public support `.dta` files, so regenerated simulated raw inputs are byte-for-byte stable across reruns.

The Figure B9 R helper sets its own R seed.

For the Fortran code, each `main.f90` sets the random-number generator to the same four fixed seed values.

Description of Programs and Execution Order

The public driver, `code/master_synth.do`, calls the core scripts in this order. The confidential driver, `code/master.do`, follows the analogous restricted-data workflow.

1. Build teacher panel without contribution measures: `oldteacher_roster_build.do`, `staff_build.do`, `teachers_build.do`, `teachers_panel.do`.
2. Build test-score and teacher-side value-added inputs: `va_teacherside_build.do`, `testscores_build.do`.
3. Estimate teacher contribution measures: `valueadded_calculation.do`, `valueadded_calculation_classmodel.do`, `valueadded_calculation_stockstaiger_nocomparativeadv.do`.
4. Construct final analysis panel: `teachers_with_allva.do`.
5. Generate model-estimation inputs: `schedule_new.do`, `gen_omegas.do`, `gen_omegas_info.do`, `build_model_dataset.do`, `build_teachers_offer_set.do`, `auxiliary_models_teachers.do`, `aux_other_moments.do`, `auxiliary_models_districts.do`, `build_model_dataset_info0.do`, `build_model_dataset_info1.do`, `build_model_dataset_info2.do`, `build_model_dataset_info3.do`, `auxiliary_models_teachers_info_matches.do`, `omega_databuild.do`, `budget_overtime.do`, `build_distance_matrix.do`, `build_wageschedule_exitentry.do`, `fix_estsample_age.do`.
6. Generate pre-Act 10 simulation inputs: `build_model_dataset_pre2011.do`, `build_simulation_1ndist.do`.
7. Generate the main-paper and appendix exhibits that do not require completed Fortran model results or the Figure B9 R helper.

Subsequently, run the Figure B9 R helper and the 31 Fortran model folders.

Lastly, `code/produce_exhibits.do` produces Figure 4, Figure B9, and the remaining generated exhibits; Table B6 is the manual exception documented in the exhibit map.

Instructions to Replicators

1. Set Up the Folder

Clone or unpack the replication package and start from the repository root.

```
cd replication
mkdir -p intermediate_data model_data simulation_data \
  out_synth/tables out_synth/figures
```

The public archive should contain `synthetic_data/raw/`, `real_data/`, and `code/`. The synthetic raw data are already included; the generator is included only so the Data Editor can inspect or regenerate them if needed.

2. Confirm the Stata Working Directory

For the public synthetic-data replication, use `code/master_synth.do`. No path edits are required. Start Stata in the repository root. The drivers set `$rootpath` from Stata's current working directory and verify the expected driver path before any build code runs. If Stata did not start there, change directories once:

```
cd ".../replication"
```

The drivers also accept being launched from the repository's `code/` directory and automatically return to the repository root. Do not change the relative structure under `synthetic_data/raw/`.

The public driver then defines the remaining globals from the detected root:

```
global code = "$rootpath/code/"
global confdata = "$rootpath/synthetic_data"
global raw = "$rootpath/real_data"
global intdata = "$rootpath/intermediate_data"
global est = "$rootpath/model_data/"
global sim = "$rootpath/simulation_data/"
global out = "$rootpath/out_synth/"
```

For a confidential-data run, use `code/master.do` and point `$confdata` to the approved local WDPI data location. Confidential data are not part of the public archive.

3. Confirm or Regenerate Synthetic Input Data

The synthetic public package should include the fictitious staff and student files listed in the “Synthetic Public Data” section above. Keep the shareable auxiliary inputs in `real_data/`.

To regenerate the simulated raw inputs from scratch, run the public simulation script from the repository root:

```
do synthetic_data/code/make_simulated_data.do
```

This recreates `synthetic_data/raw/`, refreshes `synthetic_data/documentation/synthetic_data_manifest.csv` and `synthetic_data/documentation/synthetic_data_notes.md`, and runs structural validation plus a scratch build-workflow smoke check. Run this optional regeneration only after the SSC packages checked by the master have been installed. It does not create the exhibit outputs; those are generated by the synthetic master in the next step.

4. Run the Synthetic Stata Build

From Stata:

```
do code/master_synth.do
```

From a Unix-like shell, if Stata is available on the command line:

```
stata-mp -b do code/master_synth.do
```

Replace `stata-mp` with the local Stata executable name if needed.

The synthetic run writes tables to `out_synth/tables/` and figures to `out_synth/figures/`. It also regenerates local working files in `intermediate_data/`, `model_data/`, and `simulation_data/`. The included `out_synth/` folder contains the expected outputs written directly by `code/master_synth.do`. Running the Figure B9 helper, the Fortran models, and `code/produce_exhibits.do` adds the remaining exhibits to the same folder. The public workflow should not write to `out/`.

5. Run R Simulation Script

Run the R code that produces the simulation underlying Figure B9. The helper locates the replication root from its own file path, so it can be invoked from the repository root without editing the script or changing the working directory. It installs missing packages in the configured R user library:

```
Rscript code/Figure_B9/cutoff_prefs_id_example_graph.R
```

6. Run Fortran Model Code

All of the code for the model estimation and results are located in the `code/fortran_model/` subdirectory.

Inside `code/fortran_model/` are a number of subfolders. Each subfolder contains the Fortran code to run one particular replication result; each result is subsequently saved to its own nested result subfolder.

The portable runner first verifies that `mpifort` is installed and then compiles all 31 scenarios. Compilation is the default so that invoking the script cannot accidentally begin the potentially long calculations:

```
code/fortran_model/run_all_local.sh --compile-only
```

On macOS with Homebrew, the required compiler and MPI tools can be installed with:

```
brew install gcc open-mpi
```

On Debian/Ubuntu, the corresponding system packages are commonly installed with:

```
sudo apt-get install gfortran openmpi-bin libopenmpi-dev \
  libblas-dev liblapack-dev
```

After compilation, run every scenario with an explicit MPI rank count:

```
code/fortran_model/run_all_local.sh --run --ranks <ranks>
```

On the tested 128 GB Apple M4 Max machine, eight MPI ranks are recommended. The largest executable declares approximately 6.3 GB of static data per rank, so this setting provides useful parallelism while retaining substantial memory headroom. Rank counts for other systems should be chosen according to available memory.

To compile or run only one scenario, add its path relative to `code/fortran_model/`:

```
code/fortran_model/run_all_local.sh --run --ranks <ranks> \
--scenario 1_baseline
```

The runner changes into each scenario directory before execution, as required by the relative input paths in the model code. Standard output and error are written to the scenario's `result/run_main.log`; numerical result text files remain in the same `result/` directory. Each scenario also retains its original makefile, written as a Slurm-cluster template. Its compile target requires `mpifort` plus BLAS and LAPACK:

```
make compile_mpi_slurm
```

The Slurm run target calls `sbatch run_main.batch`, but that site-specific batch file is not distributed. Use the portable runner above for a local replication or supply an appropriate batch file for a cluster. The original makefile also contains a Linux-specific library path and compiler flag; these are intentionally bypassed by the portable runner.

In total, there are 31 folders of Fortran code, each to be run in order to produce all modeling results. They are separated out and pre-set for easy replication. The scenarios are independent: none requires the generated results from another scenario as an input.

For speedy replication under synthetic data, 26 scenario folders currently set the maximum iterations and number of economies in `instruction.txt` to 2 and 2. Five folders retain the fuller settings of 1,000 iterations and 50 economies: `2_rigid`, `estimation_result`, `perturbation`, `stderr`, and `validation_result`. These five fuller settings are intentional for the public run and should not be reduced.

The original full-run configuration files are distributed under `original_fortran_instruction_files/`. For each of the 31 runnable scenarios, the relative path of the archived `instruction.txt` matches its path below `code/fortran_model/`; the Fortran source code is common to the reduced and full versions. In the 26 reduced scenarios, the archived files restore `maxiter` to 1,000, `h_maxiter` to 300, and `Neconomy` to either 10 or 50. All other operative configuration values are unchanged. To run a full configuration, copy its archived file over the corresponding public testing configuration before invoking the runner. For example:

```
cp original_fortran_instruction_files/1_baseline/instruction.txt \
code/fortran_model/1_baseline/instruction.txt
bash code/fortran_model/run_all_local.sh --ranks 8 --scenario 1_baseline
```

The runner does not select the full configurations automatically. Because they require many more iterations and simulated economies, their execution time is substantially longer than the reduced-run estimates reported above. Use a fresh package copy to return to the reduced public configurations after running a full configuration.

The fixed seeds and MPI implementation are intended to produce the same numerical results across MPI rank counts and supported compiler environments; the rank count affects computational speed rather than the model specification. The distributed Fortran `result/*.txt` files are based on the public synthetic workflow or have otherwise been cleared for public distribution.

7. Run Remaining Exhibits

While `master.do/master_synth.do` produce some exhibits, many exhibits depend on the Fortran results and are produced by another Stata file. Once all of the above results have been run, `code/produce_exhibits.do` can be run in Stata to complete the full suite of exhibits.

The Figure B10 map script leaves Stata in the `shapefile` directory. Before running the final driver, return to the replication root:

```
cd ".../replication"
do code/produce_exhibits.do
```

Exhibit Map

The entries below report the code and outputs found in this working tree. Output paths refer to the public simulated-data run under `out_synth/`. The confidential master writes its direct outputs to `out/`. The supplied `produce_exhibits.do` is configured for `out_synth/`; it must be pointed to `out/` before generating the corresponding restricted-data outputs. The Data Editor will not generate restricted-data outputs.

Main Paper Tables

Table 1: Teacher and District Characteristics (2014)

- Code/status: `code/descriptives/summary.do`
- Main input datasets: `model_data/experiencecont.dta`, `model_data/master.dta`, `model_data/contributions.dta`, `model_data/lambda.dta`, `model_data/urban.dta`, `model_data/suburban.dta`, `model_data/incumbent.dta`, `model_data/budget.dta`
- Output path: `out_synth/tables/Table_1A.log`; `out_synth/tables/Table_1B.log`

Table 2: OLS of Teacher-District Match (2014)

- Code/status: `code/auxiliary_regressions/aux_model_teachers_wages.do`
- Main input datasets: `model_data/multinomial.dta`, `model_data/matched_movers.dta`, `real_data/district_distance_sd.dta`, `real_data/cesa.dta`
- Output path: `out_synth/tables/Table_2.log`; also writes `model_data/betaT1_JPE.csv`, `model_data/betaT1_var_JPE.csv`, `model_data/betaT2_JPE.csv`, `model_data/betaT2_var_JPE.csv`

Table 3: District Wage Schedules (2014)

- Code/status: code/descriptives/summary.do
- Main input datasets: model_data/contributions.dta, model_data/experiencecont.dta, model_data/master.dta, model_data/incumbent.dta, model_data/urban.dta, model_data/suburban.dta, model_data/lambda.dta, model_data/budget.dta, model_data/alphagrid.dta, model_data/multinomial.dta; real_data/district_county_cz.dta
- Output path: out_synth/tables/Table_3A.log; out_synth/tables/Table_3B.log; out_synth/tables/Table_3C.log

Table 4: Explaining District Reward for Teacher Contribution omega_d2 (2014)

- Code/status: code/descriptives/omega_analysis.do
- Main input datasets: intermediate_data/omega_analysis_data_JPE.dta
- Output path: out_synth/tables/Table_4.log
- Upstream model-moment outputs: code/auxiliary_regressions/aux_model_teachers_wages.do writes model_data/omega1_JPE.csv, model_data/omega1_var_JPE.csv, model_data/omega2_JPE.csv, model_data/omega2_var_JPE.csv, model_data/omega_moments_JPE.csv

Table 5: Parameter Estimates

- Code/status: code/descriptives/Table_5_Table_A1.do
- Main input datasets: code/fortran_model/estimation_result/initial_param_guess.txt, code/fortran_model/stderr/result/se_param.txt
- Output path: out_synth/tables/Table_5_Table_A1.csv

Table 6: Flexible Pay vs Rigid Pay

- Code/status: code/descriptives/Table_6.do
- Main input datasets: code/fortran_model/1_baseline/result, code/fortran_model/2_rigid/result
- Output path: out_synth/tables/Table_6.csv

Table 7: State-Funded Bonuses

- Code/status: code/descriptives/Table_7.do
- Main input datasets: code/fortran_model/1_baseline/result, code/fortran_model/3_basicCfs/new1/result, code/fortran_model/3_basicCfs/new2/result, code/fortran_model/3_basicCfs/new3/result
- Output path: out_synth/tables/Table_7.csv

Table 8: Moving Costs, Resistance Costs, and Policy Impacts

- Code/status: code/descriptives/Table_8.do
- Main input datasets: code/fortran_model/4_nomovecost/case_baseline/result, code/fortran_model/4_nomovecost/case1/result, code/fortran_model/4_nomovecost/case2/result, code/fortran_model/4_nomovecost/case2_new2/result
- Output path: out_synth/tables/Table_8.csv

Main Paper Figures

Figure 1: Variation in Teacher Salaries

- Code/status: code/descriptives/data_patterns.do
- Main input datasets: intermediate_data/teachers_panel_JPE.dta, intermediate_data/indtest_math_split_JPE.dta
- Output path: out_synth/figures/Figure_1.png

Figure 2: Rates of Teacher Movements Across Districts

- Code/status: code/descriptives/data_patterns.do
- Main input datasets: intermediate_data/teachers_panel_JPE.dta
- Output path: out_synth/figures/Figure_2a.png; out_synth/figures/Figure_2b.png

Figure 3: Variation in Teacher Salaries (sorting correlations)

- Code/status: code/descriptives/data_patterns.do
- Main input datasets: intermediate_data/teachers_panel_JPE.dta, intermediate_data/indtest_math_split_JPE.dta, intermediate_data/budget_overtime_JPE.dta
- Output path: out_synth/figures/Figure_3.png

Figure 4: Cumulative Distribution Function of omega_2 by Groups of Lambda_d

- Code/status: code/descriptives/Figure_4.do
- Main input datasets: model_data/lambda.csv, model_data/alpha1.csv, model_data/alpha2.csv
- Output path: out_synth/figures/Figure_4_omega2_cdf.png

Paper Appendix A Tables

Table A1: Other Parameter Estimates: Wage Setting Cost R(omega, dem_d)

- Code/status: code/descriptives/Table_5_Table_A1.do
- Main input datasets: code/fortran_model/estimation_result/initial_param_guess.txt, code/fortran_model/stderr/result/se_param.txt
- Output path: out_synth/tables/Table_5_Table_A1.csv

Table A2: Model Fit: OLS of Teacher-District Match (post-Act 10)

- Code/status: code/descriptives/Table_A2.do
- Main input datasets: code/fortran_model/estimation_result/result/aux_and_omega_params.txt

- Output path: out_synth/tables/Table_A2.csv
- Synthetic-data note: the reduced public model makes the inferred-model regression numerically singular, so that column is missing in the distributed reference output. The remaining columns are populated.

Table A3: Model Fit: District Wage Schedules

- Code/status: code/descriptives/Table_A3_PanelA.do, code/descriptives/Table_A3_PanelB.do
- Main input datasets: code/fortran_model/estimation_result/result/aux_and_omega_params.txt, code/fortran_model/estimation_result/result/omega2_probs.txt, model_data/multinomial.dta
- Output path: out_synth/tables/Table_A3_PanelA.csv, out_synth/tables/Table_A3_PanelB.csv

Online Appendix B Tables

Table B1: Estimated parameters of teacher effectiveness

- Code/status: code/build/valueadded_calculation.do
- Main input datasets: intermediate_data/teachers_panel_nova_JPE.dta, intermediate_data/indtest_math_split_JPE.dta, intermediate_data/va_teacherside_math_JPE.dta
- Output path: out_synth/tables/Table_B1.log; also writes intermediate_data/va_highlow_math_expdisc_JPE.dta

Table B2: Distribution of teacher effectiveness

- Code/status: code/descriptives/appendix_figures.do
- Main input datasets: model_data/contributions.dta
- Output path: out_synth/tables/Table_B2.log

Table B3: Test for Forecast Unbiasedness and Non-Random Teacher Sorting

- Code/status: code/descriptives/rothstein_test.do
- Main input datasets: intermediate_data/indtest_math_split_JPE.dta, intermediate_data/teachers_panel_JPE.dta, intermediate_data/va_teacherside_math_JPE.dta
- Output path: out_synth/tables/Table_B3.tex

Table B4: Correlation of Teacher Effectiveness between Model 1 and Model 2

- Code/status: code/build/valueadded_calculation_classmodel.do
- Main input datasets: intermediate_data/teachers_panel_nova_JPE.dta, intermediate_data/va_teacherside_math_JPE.dta, intermediate_data/indtest_math_split_JPE.dta, intermediate_data/va_highlow_math_expdisc_JPE.dta
- Output path: out_synth/tables/Table_B4.log; also writes intermediate_data/va_classmodel_JPE.dta

Table B5: Auxiliary Models Aux 1a and Aux 1b, Under Achievement Models 1 and 2

- Code/status: code/auxiliary_regressions/auxmodels_teachers_varobust.do
- Main input datasets: intermediate_data/va_classmodel_JPE.dta, model_data/multinomial.dta, real_data/district_distance_sdidd.dta, real_data/cesa.dta
- Output path: out_synth/tables/Table_B5.log; also writes model_data/multinomial_varobust_JPE.dta, model_data/matched_movers_varobust_JPE.dta

Table B6: Comparison of Signal-to-Noise Ratios with Estimates of Math Teacher Value-Added in the Literature

- Code/status: No standalone replication script. The table is generated using estimates from the literature.
- Main inputs: Author calculations for the paper's value-added measures and comparisons reported in the cited literature; no separate input dataset is read by the replication workflow.
- Output path: No standalone generated file; the table is rendered as part of the online appendix.

Table B7: Sum of Squared Test Score Residuals Under c and Under (c1, c2)

- Code/status: code/descriptives/c1c2_variation_studentlevel.do
- Main input datasets: intermediate_data/teachers_panel_JPE.dta, intermediate_data/va_teacherside_math_JPE.dta, intermediate_data/va_nocomparativeadv_JPE.dta, intermediate_data/indtest_math_split_JPE.dta
- Output path: out_synth/tables/Table_B7.log

Table B8: Achievement Production Function: Controlling for Teachers and Students' Race/Ethnicity

- Code/status: code/descriptives/achievement_race.do
- Main input datasets: intermediate_data/teachers_new_JPE.dta, intermediate_data/teachers_panel_nova_JPE.dta, intermediate_data/indtest_math_split_JPE.dta, intermediate_data/va_teacherside_math_JPE.dta
- Output path: out_synth/tables/Table_B8.tex

Table B9: Cross-District Summary of Pre-Reform Wage Schedules

- Code/status: code/descriptives/schedule_summary.do
- Main input datasets: model_data/lambda.dta, intermediate_data/schedule_new_JPE.dta
- Output path: out_synth/tables/Table_B9.log

Table B10: Variation in Salaries Across and Within Districts, 2013-2016

- Code/status: code/descriptives/salary_variation.do
- Main input datasets: intermediate_data/teachers_panel_JPE.dta
- Output path: out_synth/tables/Table_B10.log

Table B11: Estimates of Aux 1a Assuming Noisy Measures of (c1, c2)

- Code/status: code/auxiliary_regressions/auxiliary_models_teachers_info.do
- Main input datasets: model_data/matched_movers.dta, model_data/multinomial.dta, model_data/matched_movers_info0-3.dta, model_data/multinomial_info0-3.dta
- Output path: out_synth/tables/Table_B11.txt

Table B12: OLS of Teacher-District Matches (Aux 1a): Baseline and Excluding Bottom-Tail Matches

- Code/status: code/auxiliary_regressions/auxiliary_models_teachers_nobottom.do
- Main input datasets: model_data/matched_movers.dta, model_data/matched_movers_nobottom.dta, model_data/multinomial.dta, real_data/district_distance_sdid.dta
- Output path: out_synth/tables/Table_B12.tex

Table B13: Teachers With One vs Many Offers: Comparison

- Code/status: code/auxiliary_regressions/auxiliary_models_teachers_richprefs.do
- Main input datasets: model_data/matched_movers.dta, model_data/multinomial.dta, real_data/district_distance_sdid.dta
- Output path: out_synth/tables/Table_B13.tex

Table B14: Parameter Permutation Exercise: Change in Estimates of Auxiliary Models from Parameter Perturbation

- Code/status: code/descriptives/Table_B14.do
- Main input datasets: code/fortran_model/perturbation/result/perturb_aux_full.txt, code/fortran_model/perturbation/result/weight_mat.txt
- Output path: out_synth/tables/Table_B14.csv

Table B15: Allocating Teachers at Will

- Code/status: code/descriptives/Table_B15.do
- Main input datasets: model_data/multinomial.dta
- Output path: out_synth/tables/Table_B15.csv

Table B16: State-Funded Teacher Bonuses: Extensive Margin

- Code/status: code/descriptives/Table_B16.do
- Main input datasets: code/fortran_model/5_exitentry/baseline/result, code/fortran_model/5_exitentry/new1_05/result, code/fortran_model/5_exitentry/new1_1/result, code/fortran_model/5_exitentry/new1_15/result, code/fortran_model/5_exitentry/new2_05/result, code/fortran_model/5_exitentry/new2_1/result, code/fortran_model/5_exitentry/new2_15/result
- Output path: out_synth/tables/Table_B16.csv

Table B17: State-Funded Teacher Bonuses: Extensive Margin and Repeated Game

- Code/status: code/descriptives/Table_B17.do
- Main input datasets: code/fortran_model/6_forward/forward_baseline/result, code/fortran_model/6_forward/forward_new1/result, code/fortran_model/6_forward/forward_new2/result
- Output path: out_synth/tables/Table_B17.csv

Table B18: State-Funded Bonuses

- Code/status: code/descriptives/Table_B18.do
- Main input datasets: code/fortran_model/1_baseline/result, code/fortran_model/3_basicCfs/new1/result, code/fortran_model/3_basicCfs/new2/result, code/fortran_model/3_basicCfs/new3/result, code/fortran_model/3_basicCfs/new4/result
- Output path: out_synth/tables/Table_B18.csv

Table B19: Program Effects With Higher Bonus Rates

- Code/status: code/descriptives/Table_B19.do
- Main input datasets: code/fortran_model/1_baseline/result, code/fortran_model/3_basicCfs/new1/result, code/fortran_model/7_higherbonus/new1_15/result, code/fortran_model/7_higherbonus/new1_2/result, code/fortran_model/7_higherbonus/new1_25/result
- Output path: out_synth/tables/Table_B19.csv

Table B20: Program Effects with $R(\cdot) = 0$

- Code/status: code/descriptives/Table_B20.do
- Main input datasets: code/fortran_model/1_baseline/result, code/fortran_model/8_noRcost/new1/result, code/fortran_model/8_noRcost/new2/result, code/fortran_model/8_noRcost/new3/result, code/fortran_model/8_noRcost/new4/result
- Output path: out_synth/tables/Table_B20.csv

Table B21: Teacher and District Characteristics (2010)

- Code/status: code/descriptives/summary_simulation.do
- Main input datasets: simulation_data/experiencecont_pre_NEW.dta, simulation_data/master_pre_NEW.dta, simulation_data/contributions_pre_NEW.dta, simulation_data/lambda_pre_NEW.dta, simulation_data/urban_pre_NEW.dta, simulation_data/suburban_pre_NEW.dta, simulation_data/incumbent_pre_NEW.dta, simulation_data/budget_NEW.dta
- Output path: out_synth/tables/Table_B21A.log; out_synth/tables/Table_B21B.log

Table B22: Model Fit: Average District Employee Characteristics

- Code/status: code/descriptives/Table_B22.do
- Main input datasets: code/fortran_model/estimation_result/result/other_aux_params.txt
- Output path: out_synth/tables/Table_B22.csv

Table B23: Model Fit: OLS of District Wage Schedule

- Code/status: code/descriptives/Table_B23.do
- Main input datasets: code/fortran_model/estimation_result/result/aux_and_omega_params.txt
- Output path: out_synth/tables/Table_B23.csv

Table B24: Model Validation: OLS of Teacher-District Match (pre-Act 10)

- Code/status: code/descriptives/Table_B24.do
- Main input datasets: code/fortran_model/validation_result/result/aux_and_omega_params.txt
- Output path: out_synth/tables/Table_B24.csv
- Synthetic-data note: the reduced public data do not identify these validation regressions, so the numeric cells are missing in the distributed reference output. This is expected for the fictitious data and does not occur in the confidential-data estimates reported in the paper.

Table B25: Model Validation: Average District Employee Characteristics (pre-Act 10)

- Code/status: code/descriptives/Table_B25.do
- Main input datasets: code/fortran_model/validation_result/result/other_aux_params.txt
- Output path: out_synth/tables/Table_B25.csv

Online Appendix B Figures

Figure B1: Distribution of Teacher Effectiveness

- Code/status: code/descriptives/appendix_figures.do
- Main input datasets: model_data/contributions.dta
- Output path: out_synth/figures/Figure_B1.png

Figure B2: Relationship Between c1 and c2

- Code/status: code/descriptives/appendix_figures.do
- Main input datasets: model_data/multinomial.dta
- Output path: out_synth/figures/Figure_B2.png

Figure B3: Relationship Between $w_d^0(x_i)$ and w_{it}^0

- Code/status: code/descriptives/appendix_figures.do
- Main input datasets: intermediate_data/teachers_panel_JPE.dta, intermediate_data/indtest_math_split_JPE.dta, intermediate_data/schedule_new_JPE.dta
- Output path: out_synth/figures/Figure_B3.png

Figure B4: Relationship Between Wage Deviations Under Two- and Three-Omega Rules

- Code/status: code/descriptives/gen_omegas_3omegas.do
- Main input datasets: intermediate_data/teachers_panel_JPE.dta, intermediate_data/schedule_new_JPE.dta, intermediate_data/indtest_math_split_JPE.dta, intermediate_data/alphagrid_wages_8groups_JPE.dta
- Output path: out_synth/figures/Figure_B4.png; also writes intermediate_data/alphagrid_wages_8groups_3omegas_JPE.dta

Figure B5: Relationship Between Wage Deviations, Untenured Teachers

- Code/status: code/descriptives/gen_omegas_tenured.do
- Main input datasets: intermediate_data/teachers_panel_JPE.dta, intermediate_data/schedule_new_JPE.dta, intermediate_data/indtest_math_split_JPE.dta, intermediate_data/alphagrid_8groups_lambdad_JPE.dta, intermediate_data/alphagrid_wages_8groups_JPE.dta
- Output path: out_synth/figures/Figure_B5.png; also writes intermediate_data/alphagrid_wages_8groups_tenured_JPE.dta

Figure B6: Relationship Between Wage Deviations, No-Experience Teachers

- Code/status: code/descriptives/gen_omegas_exp1cutoff.do
- Main input datasets: intermediate_data/teachers_panel_JPE.dta, intermediate_data/schedule_new_JPE.dta, intermediate_data/indtest_math_split_JPE.dta, intermediate_data/alphagrid_8groups_lambdad_JPE.dta, intermediate_data/alphagrid_wages_8groups_JPE.dta
- Output path: out_synth/figures/Figure_B6.png; also writes intermediate_data/alphagrid_wages_8groups_exp1cutoff_JPE.dta

Figure B7: Share of Teachers Who Switch In and Out of Math Teaching, By Year

- Code/status: code/descriptives/budget_shifts.do
- Main input datasets: intermediate_data/teachers_new_JPE.dta
- Output path: out_synth/figures/Figure_B7.png

Figure B8: Share of Districts' Budgets Spent on Teacher Salaries, by Grade and Subject

- Code/status: code/descriptives/budget_shifts.do
- Main input datasets: intermediate_data/teachers_new_JPE.dta
- Output path: out_synth/figures/Figure_B8A.png; out_synth/figures/Figure_B8B.png; out_synth/figures/Figure_B8C.png; out_synth/figures/Figure_B8D.png

Figure B9: Identification Illustrated

- Code/status: code/Figure_B9/Figure_B9.do; helper code/Figure_B9/cutoff_prefs_id_example_graph.R creates the input
- Main input datasets: code/Figure_B9/cutoffs_id_exercise.dta
- Output path: out_synth/figures/Figure_B9_b_02.png; out_synth/figures/Figure_B9_b_1.png; out_synth/figures/Figure_B9_b_5.png

Figure B10: Distribution of λ Across Wisconsin Districts

- Code/status: code/descriptives/lambda_map.do
- Main input datasets: real_data/district_shapefiles/WI_School_Districts.*, intermediate_data/indtest_math_split_JPE.dta, model_data/lambda.dta
- Output path: out_synth/figures/Figure_B10.png

Data References

Wisconsin Department of Public Instruction. 2006–2016. Wisconsin administrative staff, roster, student-demographic, and test-score records [restricted-use data]. Madison, WI. Data first accessed November 2017 under a Data User Agreement; WDPI did not assign a formal version or release number. Access information: “Confidential Data Application.” <https://dpi.wi.gov/wise/data-requests/confidential-data-application>.

Wisconsin Department of Public Instruction. n.d. “Cooperative Educational Service Agencies.” <https://dpi.wi.gov/about-dpi/cesa>. Used by the authors to construct real_data/cesa.dta.

Wisconsin Department of Public Instruction. 2024. “School Districts, Wisconsin.” Edition 8.0, revised June 26, 2024; valid for the 2024–2025 school year. Downloaded in 2025 from “GIS Maps.” <https://dpi.wi.gov/wisedash/public/gis-maps>. Provider metadata are included in real_data/district_shapefiles/WI_School_Districts.shp.xml.

National Center for Education Statistics and U.S. Census Bureau. n.d. “American Community Survey–Education Tabulation (ACS-ED), 2013–2017: School-District Profile Tables.” U.S. Department of Education, Institute of Education Sciences. <https://nces.ed.gov/programs/edge/demographic/acsedtables>. The original download date was not recorded. Provider record layouts and ReadMe files are included beside each extract under real_data/nces_acs_2013–2017/.

National Center for Education Statistics. n.d. “School District Boundaries.” Education Demographic and Geographic Estimates (EDGE), U.S. Department of Education, Institute of Education Sciences. <https://nces.ed.gov/programs/edge/Geographic/DistrictBoundaries>. Used to prepare real_data/crosswalk_census_state_id.csv; the exact source vintage was not recorded.

Bergeron, Augustin, Raj Chetty, David Cutler, Benjamin Scuderi, Michael Stepner, and Nicholas Turner. 2022. “Replication Data for: The Association Between Income and Life Expectancy in the United States, 2001–2014.” Harvard Dataverse, V1. <https://doi.org/10.7910/DVN/VVW76J>. In particular, the authors used the dataset’s cty_cz_st_crosswalk county-commuting-zone-state crosswalk to prepare the district and commuting-zone crosswalks distributed here.

Daily Kos Elections. 2021. “Daily Kos Elections’ Statewide Election Results by Congressional and Legislative Districts.” March 23, 2021. <https://www.dailykos.com/stories/2013/07/09/1220127/-Daily-Kos-Elections-2012-election-results-by-congressional-and-legislative-districts>. Accessed in 2022.

Daily Kos Elections. n.d. “Daily Kos Elections Statewide Election Results by Congressional Districts Used from 2012–2020.” Google Sheets. <https://docs.google.com/spreadsheets/d/17yr9mcAtuUdNjI9NEPYKxXsEldzzQ2ZaDwEAbnPRyS4/>. Accessed in 2022.

Daily Kos Elections. n.d. “Daily Kos Elections Counties–Congressional District Overlaps (117th Congress).” Google Sheets. <https://docs.google.com/spreadsheets/d/18adZplghSQQTZLrUNzEdn78ng7mnk2l4-h6lYPsv34l/>. Accessed in 2022.

Google LLC. 2022. “Google Maps.” <https://www.google.com/maps>. Accessed in September 2022 and used by the authors to construct real_data/district_distance_sdidd.dta.

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